Given $y(t)=x(t)*h(t)$, we want $y(t)=\sum_k(\dots)e^{jkw_0t}$, what is $(\dots)$?

Consider $x(t)=\sum_{k=-\infty}^{\infty}a_ke^{jk\omega_0t}$, a_k is a scaler, then

$$y(t)=x(t)*h(t)=\left(\sum_{k=-\infty}^{\infty}a_ke^{jk\omega_0t}\right)*h(t)=\sum_{k=-\infty}^{\infty}a_k(e^{jk\omega_0t}*h(t)).$$

For single $e^{jk\omega_0t}*h(t)$, we have

$$e^{jk\omega_0t}*h(t)=\int_{-\infty}^{\infty}h(\tau)e^{jk\omega_0(t-\tau)}\mathrm{d}\tau=e^{jk\omega_0t}\int_{-\infty}^{\infty}h(\tau)e^{-jk\omega_0\tau}\mathrm{d}\tau.$$

Let $H(j\omega_k)=\int_{-\infty}^{\infty}h(\tau)e^{-jk\omega_0\tau}\mathrm{d}\tau$, then

$$y(t)=\sum_{k=-\infty}^{\infty}a_k(e^{jk\omega_0t}*h(t))=\sum_{k=-\infty}^{\infty}a_kH(j\omega_k)e^{jk\omega_0t}.$$

For a given $h(t)$, $H(j\omega_k)$ is easy to calculate.

So the problem becomes: **when there is a $x(t)$, how to find its a_k for each e^{jkw_0t}?**

Consider multiplying $x(t)$ by e^{-jnw_0t}, we have

$$x(t)e^{-jnw_0t}=\sum_{k=-\infty}^{\infty}a_ke^{jkw_0t}e^{-jnw_0t}=\sum_{k=-\infty}^{\infty}a_ke^{j(k-n)w_0t}.$$

Then

$$\int_0^T x(t)e^{-jnw_0t}\mathrm{d}t=\int_0^T\sum_{k=-\infty}^{\infty}a_ke^{j(k-n)w_0t}\mathrm{d}t=\sum_{k=-\infty}^{\infty}a_k\left[\int_0^T e^{j(k-n)w_0t}\mathrm{d}t\right].$$

For single $\int_0^T e^{j(k-n)w_0t}\mathrm{d}t$, we have

$$\int_0^T e^{j(k-n)w_0t}\mathrm{d}t=\int_0^T\cos((k-n)w_0t)\mathrm{d}t+j\int_0^T\sin((k-n)w_0t)\mathrm{d}t=\begin{cases}T, & k=n\\ 0, & k\neq n\end{cases}.$$

Thus,

$$\int_0^T x(t)e^{-jnw_0t}\mathrm{d}t=Ta_n\Longrightarrow a_n=\frac{1}{T}\int_0^T x(t)e^{-jnw_0t}\mathrm{d}t\Longrightarrow\boxed{a_k=\frac{1}{T}\int_T x(t)e^{-jkw_0t}\mathrm{d}t}.$$

Now a_k and $H(j\omega_k)$ are both computable, then we can find $y(t)$ by $\sum_{k=-\infty}^{\infty}a_kH(j\omega_k)e^{jk\omega_0t}$.

When $x(t)$ is real,

$$\boxed{a_k = a_{-k}^*}.$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k\, e^{jkw_0 t} = a_0 + \sum_{k=1}^{\infty} 2|a_k| \cos(k w_0 t + \arg(a_k)).$$

When $x(t)$ is odd,

$$\begin{aligned} a_{-k} &= \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{jk\omega_0 t}\, \mathrm{d}t \\ &= \frac{1}{T}\int_{T/2}^{-T/2} x(-t')\, e^{-jk\omega_0 t'}\, \mathrm{d}(-t') \\ &= \frac{-1}{T}\int_{-T/2}^{T/2} x(t')\, e^{-jk\omega_0 t'}\, \mathrm{d}t' \\ &= -a_k. \end{aligned}$$

So

$$\mathrm{Re}(a_k) = \mathrm{Re}(a_{-k}) = 0.$$

When $x(t)$ is even,

$$\begin{aligned} a_{-k} &= \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{jk\omega_0 t}\, \mathrm{d}t \\ &= \frac{1}{T}\int_{T/2}^{-T/2} x(-t')\, e^{-jk\omega_0 t'}\, \mathrm{d}(-t') \\ &= \frac{1}{T}\int_{-T/2}^{T/2} x(t')\, e^{-jk\omega_0 t'}\, \mathrm{d}t' \\ &= a_k. \end{aligned}$$

So

$$\mathrm{Im}(a_k) = \mathrm{Im}(a_{-k}) = 0.$$

	CT	DT
angular freq	$w_0 = {}^{2\pi}/_{T_0}$	$\Omega_0 = {}^{2\pi}/_{N_0}$
FS rep of x	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$	$x[n] = \sum_{k=\langle N_0\rangle} a_k e^{jk\Omega_0 n}$
FS rep of y	$y(t) = \sum_{k=-\infty}^{\infty} a_k H(j\omega_k) e^{jk\omega_0 t}$	$y[n] = \sum_{k=-\infty}^{\infty} a_k H(j\omega_k) e^{jk\Omega_0 n}$
FS response	$H(j\omega_k) = \int_{-\infty}^{\infty} h(\tau) e^{-jk\omega_0 \tau} \mathrm{d}\tau$	$H(j\omega_k) = \sum_{n=-\infty}^{\infty} h[n] e^{-jk\Omega_0 n}$
coefficients of x	$a_k = \frac{1}{T_0}\int_{T_0} x(t) e^{-jkw_0 t} \mathrm{d}t$	$a_k = \frac{1}{N_0}\sum_{n=\langle N_0\rangle} x[n] e^{-jk\Omega_0 n}$
coefficients of y	$b_k = a_k H(j\omega_k)$	$b_k = a_k H(j\omega_k)$

Table 1. Summary & Comparision